Name : …… Student ID: …….

The University of New South Wales

School of Electrical Engineering and Telecommunications

Mid-term Examination (Morning)

Term 3, 2021

ELEC3104: Digital Signal Processing

Candidates should attempt the questions as per the following Instructions

1. This mid-term exam constitutes 20% of the total course assessments
2. **Time allowed:** 2 hours (+ 10 min reading time + 20 minutes uploading answers)
3. You must answer a maximum of 5 questions (including the MATLAB Question)
4. **Except** the MATLAB question, all other questions are presented in the SOLO Modified Taxonomy form:
   - The exam will be in the following layout:
     - ❖ Four questions under SOLO - Level 2
     - ❖ One question under SOLO - Level 3
     - ❖ One question under SOLO - Level 4
     - ❖ One question under SOLO - Level 5
   - In doing the exam, you can choose which questions you want to complete, depending on what Level you are aiming for:
     - ❖ If you are aiming to achieve **Level 2**, you will need to **answer all four questions under Level 2** + the MATLAB question. The maximum mark you can achieve is **64.**
     - ❖ If you are aiming to achieve **Level 3**, you will need to answer **any three Level 2 questions and one Level 3 question** + the MATLAB question. The maximum mark you can achieve is **74**.
     - ❖ If you are aiming to achieve **Level 4**, you will need to answer **any two Level 2 questions, one Level 3 question and one Level 4 question** + the MATLAB question. the maximum mark you can achieve is **84**.
     - ❖ If you are aiming to achieve **Level 5**, you will need to **answer any one of the Level 2 questions, one Level 3 question, one Level 4 question and one Level 5 question +** the MATLAB question. The maximum mark you can achieve is **100**.
5. Print your name, student ID and question number on the front page of each **answer** page.
6. The questions are NOT of equal value. Part marks are as indicated.
7. Authorised examination materials: This is an open book exam.
8. This paper contains 11 pages apart from this instruction page.

ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY ONLY BE USED FOR DRAWING, SKETCHING OR GRAPHICAL WORK. UPLOAD AN IMAGE OF EACH ANSWER INDIVIDUALLY ON MOODLE WITHIN THE ALLOCATED TIME

SOLO – LEVEL 2

If you are aiming to achieve **Level 2**, you will need to **answer all four questions under Level 2** + the MATLAB question. The maximum mark you can achieve is **64.**

SOLO – Level 2: Question 1 (13 marks)

(a) An analogue signal x(t) = 3cos(2000πt) + 5sin (6000πt) + 10cos(12000πt) is sampled 5000 times per second. How many frequency components will the discrete-time signal obtained after sampling contain? Write down the expression for this discrete-time signal.

[4 marks]

(b) Given a real-valued bandpass signal x(t) with $X(f) = 0$ for $|f| < f_1$ and $|f| > f_2$ (see Figure 1), the Nyquist sampling theorem says that the minimum sampling frequency is $f_s = 2\ f_2$. However, in some cases, the signal may be sampled at a lower rate. Show that no aliasing will occur if $x(t)$ is sampled at a sampling frequency $f_s = 2\ (f_2 - f_1)$; where $f_2 = 10$ kHz and $f_1 = 8$ kHz. Sketch the magnitude response of the digital spectrum $X(\theta)$.

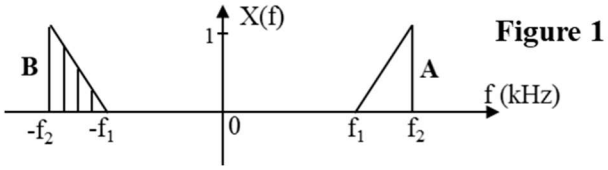
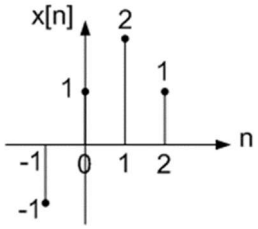


Figure 1

[5 marks]

(c) A discrete time signal x[n] is shown below. Sketch $y[n] = x[-n-1]$. **[4 marks]**



SOLO – Level 2: Question 2 (13 marks)

(a) A first-order digital low-pass filter is described by the system function

$$H(z) = \frac{k}{1 - az^{-1}} \qquad 0 < a < 1$$

(i) Determine k, so that the maximum value of |H(θ)| is equal to 1. **[2 Marks]**

(ii) The 3dB cut-off frequency of the transfer function $H(\theta)$ is θ_c. Show that a is given by

$$a = (2 - \cos\theta_c) - \sqrt{(2 - \cos\theta_c)^2 - 1}$$ **[5 marks]**

(b) A digital filter is represented by,

$$\boldsymbol{H(z) = \frac{\frac{1}{2}}{1 - \frac{1}{2}z^{-1}} + \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)(1 - z^{-1})}}$$

Implement a parallel realization of H(z) using only two delay elements.

[6 marks]

SOLO – Level 2: Question 3 (13 marks)

(a) Determine the coefficients of a linear -phase filter (given below) such that its frequency response is normalised so that H(0) = 1.

$$\boldsymbol{y[n] = a_0\, x[n] + \frac{1}{3}\, x[n-1] + a_2 x[n-2]}$$

[4 Marks]

(b) Determine the transfer function H(z) of a discrete - time system with the pole-zero map shown in Figure 2:

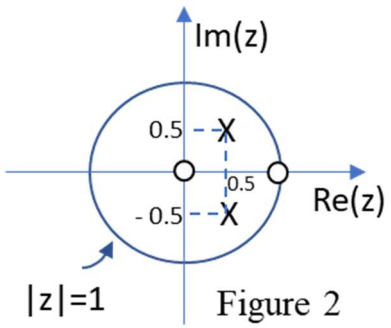


Figure 2

[4 Marks]

(c) Using the pole-zero map shown in part (b) above, sketch an approximate magnitude response of the H(z) **[5 Marks]**

SOLO – Level 2: Question 4 (13 marks)

(a) Determine and sketch the phase response of the following filter:

$$y[n] = x[n] - x[n-1]$$ **[4 Marks]**

(b) (i) Write the equation for the 4-point DFT, H[k], of the impulse response of a filter h[n] = [1, 2, 3]. **[3 marks]**

(ii) Find the value of H[3] as a complex number of cartesian form, based on (i) above. **[2 marks]**

(iii) What is the DC gain of this filter, based on (i) above? **[2 Marks]**

(iv) Find the value of H[1], using the properties of DFT without explicitly calculating it. Explain your answer. **[2 Marks]**

End of SOLO - Level 2 Examination Questions.

SOLO – LEVEL 3

If you are aiming to achieve **Level 3**, you will need to answer **any three Level 2 questions and one Level 3 question** + the MATLAB question. The maximum mark you can achieve is **74**.

SOLO – Level 3: Question (23 marks)

(a) The frequency response $H_d(\theta)$ of a linear phase FIR filter is given by:

$$H_d(\theta) = \begin{cases} e^{-j3\theta} & 0 \leq |\theta| \leq \pi/2 \\ 0 & elsewhere \end{cases}$$

(i) Show that the impulse response $h_d[n]$ is given by

$$h_d[n] = \frac{sin\left(\frac{\pi}{2}(n-3)\right)}{\pi(n-3)}$$

[7 marks]

(ii) The length of the impulse response of this linear phase FIR filter is set to 7. Calculate $h_d[n]$ using part (i) above, for $n = 0, 1, 2, 3$.

[3 marks]

(iii) The coefficients are symmetrical about n = 3. Find the values of all 7 filter coefficients.

[3 marks]

(iv) Using minimum number of multiplications, draw an implementation (structure) of the FIR filter obtained in (iii) above.

[4 marks]

(b) The system function of a causal FIR filter is:

$$H(z) = (\mathbf{1 - 0.3\ z^{-1}} + 0.9\ \mathbf{z^{-2}})(1 - \mathbf{0.5\ z^{-1}})$$

Find three other causal FIR filters with h[0] > 1 that have frequency response with the same magnitude.

[6 marks]

End of SOLO - Level 3 Examination Question.

SOLO – LEVEL 4

If you are aiming to achieve **Level 4**, you will need to answer **any two Level 2 questions, one Level 3 question and one Level 4 question** + the MATLAB question. the maximum mark you can achieve is **84**.

SOLO – Level 4: Question 1 (23 marks)

(a) Consider the interconnection of three LTI systems as shown in Figure 3:

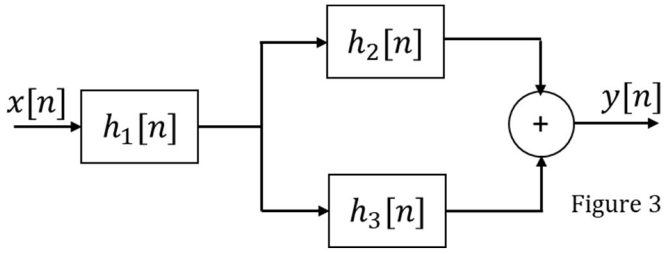


Figure 3

If $h_1[n] = u[n-2]$, $h_2[n] = n\,u[n]$ *and* $h_3[n] = \delta[n-2]$, find the unit sample (impulse) response of the overall system.

[9 marks]

(b) Show that the system specified by the pole-zero plot in Figure 4 is an all-pass filter.

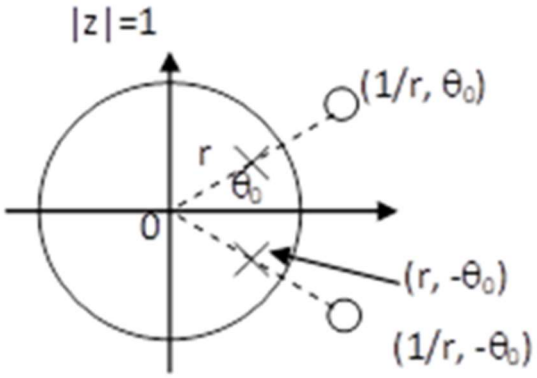


Figure 4

[9 marks]

(c) Assuming that the digital filter G(z) is to be realised using a cascade structure, draw a suitable realisation diagram.

$$G(z) = \frac{(z+1)^3}{6z\ (z - j\frac{1}{\sqrt{3}})\ (z + j\frac{1}{\sqrt{3}})}$$

[5 marks]

End of SOLO - Level 4 Examination Question.

SOLO – LEVEL 5

If you are aiming to achieve **Level 5**, you will need to **answer any one of the Level 2 questions, one Level 3 question, one Level 4 question and one Level 5 question +** the MATLAB question. The maximum mark you can achieve is **100**.

SOLO – Level 5: Question (29 marks)

(a) An FIR linear phase filter has a unit sample response (impulse response) that is real with $h[n] = 0$ for $n < 0$ and $n > 7$. If $h[0] = 1$ and the system function has a zero at $z = 0.4\, e^{j\pi/3}$ and a zero at z=3, what is H(z)?

[10 marks}

(b) The Frequency response H(θ) of a filter is shown in Figure 5. Find the impulse response $h[n]$.

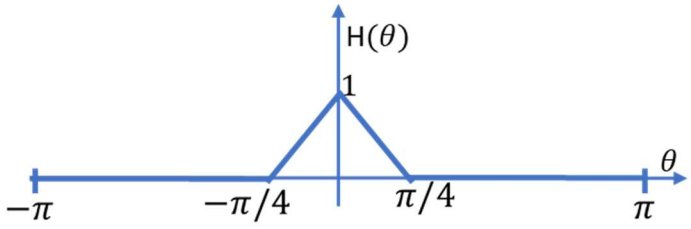


Figure 5

[10 marks]

(c) Find a real-valued causal sequence $x[n]$ with $x[0] > 0$ and $|X(\theta)|^2 = 1 + a^2 - 2a\cos\theta$, where $0 < a < 1$.

[9 marks]

End of SOLO - Level 5 Examination Question.

MATLAB Question

You must answer the MATLAB question.
You should save your program in the file *midexam_zID.m* (zID is your student zID) for uploading.

SOLO – Level 1: Question (12 marks)

A discrete-time signal 's' saved in a wav file '*sMe.wav'*, located in folder 'Wav file' (Week 5 folder on Moodle). Write a MATLAB program to read the wav file to obtain the signal 's' and its sampling rate fs and then do the following tasks:

1. Filter the signal 's' by the filter $H(z) = \frac{z^2+0.9801}{z^2+0.81}$ to obtain the signal 'y' **[2 marks]**
2. Plot the first 400 samples of the signals 's' and 'y'. The horizontal axis should be in milli second. **[2 marks]**
3. Compute and plot the magnitude spectrum of the signals 's' and 'y' in the frequency region [0, fs/2]. The horizontal axis should be in Hertz. **[3 marks]**
4. Determine the poles & zeros of the filter and plot the pole-zero plane (zplane) **[3 marks]**
5. Compute and plot the magnitude response of the filter H(z). The horizontal axis should be in Hertz. **[2 marks]**

Your program should plot all requested figures with the appropriate label on each axis including a title.

Hint: The in-built functions *audioread* could be used to read the wav file and its sampling rate